

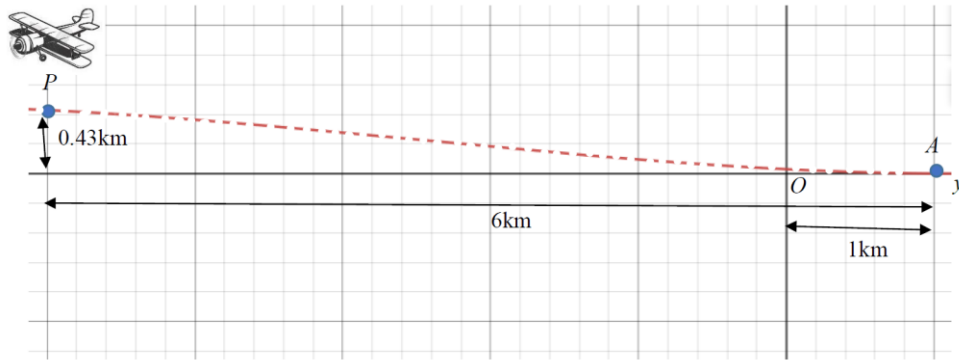
**QUESTION ONE**

- (a) Simplify the expression, leaving your answer with positive indices  $\left(\frac{2x^3y^2}{x^2}\right)^{-2}$
- (b) If  $(2 - 3x) \times f = 15x^2 - 22x + 8$   
Find  $f$  in terms of  $x$  and express it in its simplest form.
- (c) If  $x = \log 2$  and  $y = \log 5$ , write  $\log \sqrt[3]{10}$  in terms of  $x$  and  $y$ .
- (d) Make  $x$  the subject of the formula  $p = \sqrt{\frac{2-ax^2}{x^2-b}}$
- (e) The following quadratic equation, where  $m$  is a constant, has two distinct real roots.  
$$x^2 + (m + 2)x + 4m - 7 = 0$$
  
Determine the range of possible values for  $m$ .
- (f) Given that  $a = x^{\frac{1}{2}} + x^{-\frac{1}{2}}$  and  $b = x^{\frac{1}{2}} - x^{-\frac{1}{2}}$   
Show that  $a^2b^2 + 4 = \left(x + \frac{1}{x}\right)^2$

**QUESTION TWO**

- (a) Write as a single logarithm in simplest form:  $\log x^3 + \log \frac{1}{x} - 5 \log x$
- (b) A function is defined as  $f(x) = x^2 - 10x + 4$   
Express  $f(x)$  in completed square form, i.e.  $f(x) = (x + a)^2 + b$ , where  $a$  and  $b$  are integers.
- (c) A car purchased new for \$52,000 in January 2014 depreciates at a rate of 20% per year.  
Its value  $V$  after  $t$  years, in thousands of dollars, can be modelled by:  $V = 52 \times 0.8^t$ .
- (i) Use algebra to find after how many years was the car worth \$24,000?
- (ii) A second car, initially worth \$39,500 in January 2014 loses value at 15% per year.  
In what year were the two cars worth the same? Show all algebraic working.

- (d) A plane is flying at a height of 0.43km above ground level when it begins its descent 6km from the airport at point A. The airport is 1km from the origin O.

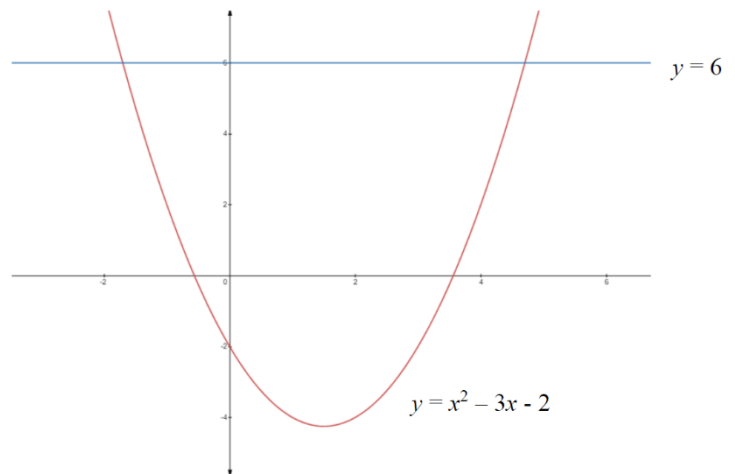


Taking O as the origin, the curve (shown above as a dotted line) approximately describes the path of the plane's descent where  $f(x) = 0.0024x^3 + 0.018x^2 - cx + d$ .

Use algebra to find the value of  $c$  and  $d$  and hence state the full function which approximates the plane's descent.

### QUESTION THREE

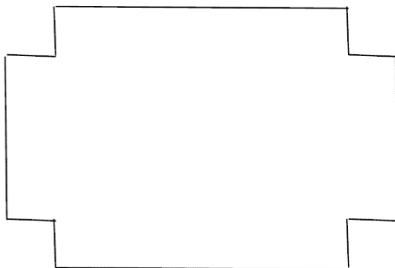
- (a) Simplify  $\frac{x}{x+5} + \frac{2}{x-3}$
- (b) Solve.  $\frac{5x^2-72}{x-3} = 2x$
- (c) Find the nature of the roots of  $7a^2 + 5(a+1) = 3$
- (d) Use algebra to find the **exact**  $x$  values where the parabola  $y = x^2 - 3x - 2$  crosses the line  $y = 6$  as shown.



- (e) A rectangular piece of sheet metal is 7 cm shorter than it is wide.

From each corner, a  $3 \times 3$  cm square is cut out and the flaps are then folded up to form an open box, as shown below.

If the volume of the box is  $1620 \text{ cm}^3$ , find the length and width of the original piece of sheet metal.



- (f) The quadratic equation  $3(k+2)x^2 - (5k+7)x + 3k+1 = 0$  where  $k$  is a constant and  $k \neq -2$ , has two distinct real roots.

Use algebra to justify that  $\frac{-25}{11} < k < 1$